Geometry & Trigonometry

ANSWER KEY

Area and perimeter of basic shapes

1 Find the area of a triangle with base 12 and height 7.
Area $= \frac{1}{2}(12)(7) = 42$.

2 Find the area and circumference of a circle with radius 5, in terms of π.
Area $= \pi r^2 = 25\pi$. Circumference $= 2\pi r = 10\pi$.

The Pythagorean theorem

3 A right triangle has legs 6 and 8. Find the length of the hypotenuse.
$c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

4 A ladder 13 feet long leans against a wall, with its base 5 feet from the wall. How high up the wall does the ladder reach?
Using the Pythagorean theorem: $h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ feet.

Volume and surface area of solids

5 Find the volume of a rectangular prism with dimensions 4 by 5 by 6.
Volume $= 4 \times 5 \times 6 = 120$ cubic units.

6 Find the volume of a cylinder with radius 3 and height 10, in terms of π.
Volume $= \pi r^2 h = \pi(9)(10) = 90\pi$.

Similar triangles and proportional reasoning

7 Triangle ABC is similar to triangle DEF, with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF.
The scale factor is $\frac{DE}{AB} = \frac{9}{6} = 1.5$. So $EF = 1.5 \times BC = 1.5 \times 8 = 12$.

8 A tree casts a shadow 20 feet long, while a 6-foot person standing nearby casts a shadow 4 feet long. Find the height of the tree using similar triangles.
Set up a proportion: $\frac{\text{tree height}}{20} = \frac{6}{4}$. So tree height $= \frac{6}{4} \times 20 = 30$ feet.

Right triangle trigonometry

9 In a right triangle, the angle θ has an opposite side of length 5 and hypotenuse 13. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

The adjacent side is $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin\theta = \frac{5}{13}$, $\cos\theta = \frac{12}{13}$, $\tan\theta = \frac{5}{12}$.

10 A 20-foot ladder leans against a wall, making a 60° angle with the ground. Find the height the ladder reaches on the wall. to the nearest tenth.

Height $= 20\sin(60^\circ) = 20\left(\frac{\sqrt{3}}{2}\right) = 10\sqrt{3} \approx 17.3$ feet.

Angle relationships and circle theorems

11 Two parallel lines are cut by a transversal. If one angle measures 65°, find the measure of its corresponding angle and its co-interior (same-side interior) angle.

The corresponding angle is equal: 65°. The co-interior angle is supplementary to it (they sum to 180°): $180^\circ - 65^\circ = 115^\circ$.

12 A central angle in a circle measures 80°. Find the measure of the inscribed angle that subtends the same arc.

An inscribed angle is always half the central angle subtending the same arc: $\frac{80^\circ}{2} = 40^\circ$.

The unit circle and radian measure

13 Convert 120° to radians, in terms of π.

$120^\circ \times \frac{\pi}{180^\circ} = \frac{2\pi}{3}$.

14 Find the exact value of $\cos\left(\frac{\pi}{3}\right)$ using the unit circle.

$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$.

Applying geometry and trigonometry to real-world contexts

15 A surveyor stands 100 feet from the base of a building and measures the angle of elevation to the top as 35°. Find the height of the building, to the nearest foot.

Height $= 100\tan(35^\circ) \approx 100(0.7002) \approx 70$ feet.

Composite area and shaded regions

16 A square with side length 10 has a circle of radius 4 cut out of its center. Find the remaining area, in terms of π.

Square area: $10^2 = 100$. Circle area: $\pi(4)^2 = 16\pi$. Remaining area: $100 - 16\pi$.

17 Find the area of a trapezoid with parallel sides 6 and 10, and height 5.
Area $= \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(6 + 10)(5) = \frac{1}{2}(16)(5) = 40.$

Coordinate geometry: distance and midpoint

18 Find the distance between the points $(1, 2)$ and $(7, 10)$.
Distance $= \sqrt{(7-1)^2 + (10-2)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$

19 Find the midpoint of the segment connecting $(-3, 5)$ and $(9, -1)$.
Midpoint $= \left(\frac{-3+9}{2}, \frac{5+(-1)}{2}\right) = \left(\frac{6}{2}, \frac{4}{2}\right) = (3, 2).$

Equation of a circle

20 Write the equation of a circle with center $(2, -3)$ and radius 5.
$(x-2)^2 + (y+3)^2 = 25.$

21 Find the center and radius of the circle $(x+1)^2 + (y-4)^2 = 36.$
Center: $(-1, 4).$ Radius: $\sqrt{36} = 6.$

Arc length and sector area

22 A circle has radius 6. Find the arc length of a 60° sector, in terms of π.
Arc length $= \frac{\theta}{360^\circ} \times 2\pi r = \frac{60}{360} \times 2\pi(6) = \frac{1}{6} \times 12\pi = 2\pi.$

23 Find the area of a 90° sector of a circle with radius 8, in terms of π.
Sector area $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{90}{360} \times \pi(64) = \frac{1}{4} \times 64\pi = 16\pi.$